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Behavioural addictions in obsessive compulsive disorder – Prevalence and clinical correlates

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ABSTRACT

Behavioural addictions (BAs) share phenomenological overlap and neuropsychological similarities with obsessive-compulsive disorder (OCD); however, the relationship remains understudied.

This study reports the prevalence of BAs in patients with OCD and healthyxcontrols (HCs). It compares the clinical characteristics including impulsivity traits among OCD subjects with and without BAs. A total of 150 OCD patients and 131 HCs were assessed using structured instruments for the following BAs (Internet/Sex/Pornography/Mobile phone/Kleptomania/Compulsive buying/Pyromania/Gambling/Food/Exercise). Impulsivity was assessed using Barratt Impulsiveness Scale (BIS-11).

Subjects: with OCD reported significantly higher prevalence of atleast one co-occurring BA compared to HC [OCD (70%); HC (58%), $p < 0.036$]. They also reported a significantly higher prevalence of Internet Addiction (29.3% vs 3.1%), pornographic addiction (12.7% vs 0%), and sex addiction (10% vs 1.5%). Though use of mobile phone use and food addiction were not significantly different when compared using the diagnostic criteria, subjects with OCD reported significantly higher intensity of symptoms when comparison was made using severity scores.

When OCD patients with and without BAs were compared, those with BAs were significantly male, with a lower age at onset and assessment. They had significantly higher total impulsivity scores, motor and attentional subscale scores on BIS. The two groups did not differ in symptom profile and comorbidities of OCD.

To conclude, the co-occurrence of BAs among subjects with OCD is common. Additionally, OCD subjects with co-occurring BAs have certain specific demographic characteristics and higher impulsivity.

1. Introduction

Behavioural addictions (BAs) are conceptualised as disorders characterised by inability to control intense urges resulting in persistent repetitive behaviour despite significant negative impacts on personal, social and occupational functioning (Kardefelt-Winther et al., 2017). BAs shares phenomenological similarities with obsessive-compulsive disorder (OCD). Common are the recurrent, persistent thoughts, images and need to perform repetitive behaviour to reduce distress, despite recognising their behaviour to be excessive or unreasonable. Though features appear to be shared, BAs are largely ego-syntonic meaning that most often behaviours are preceded by "gratification, pleasure or relief" rather than the ego-dystonic compulsive behaviours in OCD meant to reduce anxiety and distress of obsessive thoughts. However, with chronicity, addictive behaviours become less pleasurable, habitual and ego-dystonic. Conversely, in those with childhood OCD or with poor insight, ego-dystonic compulsions may not be the rule (El-Guebaly et al., 2012)(Frost et al., 1998)(Ruscio et al., 2010)(Stein, 2008). In addition to phenomenological overlap, those with OCD and gambling disorder display increased impulsivity, impaired decision-making, and reward system dysfunction (Grassi et al., 2020). Both habitual and affective motivations have been reported to underly both OCD and alcohol use disorder albeit in differing proportions (Ferreira et al., 2021). Cortico-striatal dysfunction in the partially overlapping circuits which underly OCD and Impulse control disorders has been postulated to explain these shared commonalities (Fontenelle et al., 2011).

Despite this phenomenological overlap and neuropsychological data, the relationship between OCD and BA is still understudied. A clarity on the relationship may ultimately improve the conceptualization of BA along the compulsive, impulsive and addiction spectrum (Stein, 2008). Clinically, a relationship between two disorders is best suggested by evidence of over-representation of disorders when systematically

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examined. Most studies have focussed on the relationship between BA and substance use disorders, studies on OCD are relatively sparse (Grant and Chamberlain, 2020)(Starcevic and Aboujaoude, 2017).

In the largest study to date on BAs in OCD across 13 countries, among 6916 OCD subjects, 8.7% reported problematic internet use, 6.8% compulsive sexual behaviour disorder, 6.4% compulsive buying and 4.1% gambling. However, most centres diagnosed BAs clinically, with no diagnostic screening instruments being applied and only four disorders deemed common across various centres being examined (Brakoulias et al., 2020). Other studies have reported the relationship of OCD or OCD symptoms (OCS) in a single BA. The findings suggest that there are elevated rates of OCD/OCS in internet addiction (Bipeta et al., 2015), compulsive buying (Lejoyeux et al., 2005), pathological gambling (Durdle et al., 2008) and compulsive sexual behaviours (Fuss et al., 2019).

Existing studies are thus limited in that they have only 'clinically examined' multiple BAs or systematically examined only a single BA in OCD. A more clinically meaningful conceptualization on the relationship between OCD and BAs will be better served by examining all disorders currently proposed as BA systematically rather than pre-selecting a particular set of disorders. There is robust evidence of OCD being closer to the compulsivity end of the impulsive-compulsive spectrum with high harm avoidance (Fontenelle et al., 2021). However, recent evidence suggests a sub-group of OCD patients may have higher impulsivity traits (Kashyap et al., 2012). Given that subjects with BAs report high levels of impulsivity, it is heuristically attractive to examine impulsivity traits among subjects with OCD as a possible phenotypic marker of its relationship with BA. Additionally, given that OCD is heterogenous, it's unclear at this point whether only a sub-group of OCD subjects have a higher risk of BA. Understanding this would require comparing the clinical characteristics of OCD subjects with and without BA. In this background, our study aimed to examine the prevalence of BAs in patients with OCD and compare them with healthyxcontrols (HC). Further, it will compare the socio-demographics characteristics, clinical features including impulsivity traits and the mean comorbidities among OCD subjects with and without BA. Additionally, subjects with OCD with BA will show distinctive clinical characteristics as compared to those without comorbid BA. We have also hypothesised that impulsivity would be higher among those with who have comorbid BAs and OCD compared to those with OCD alone and healthy controls.

2. Methods

2.1. Participants

The study is a cross-sectional survey of adult subjects (age 18–60 years) with a primary diagnosis of OCD newly registered at the specialty obsessive-compulsive disorder clinic of the National Institute of Mental Health & Neurosciences, Bengaluru, India from September 2017 to November 2018. Subjects were excluded if they had co-morbid bipolar disorder, psychotic disorders, substance dependence (except nicotine), organic brain disorders, intellectual developmental disorder or any pervasive developmental disorder. The speciality OCD Clinic provides services every Tuesday. During the period, there were 320 new registrations, all of whom were invited to participate. Among them, 38 did not fulfil the inclusion criteria, 112 refused consent and assessments were completed for 150 patients (response rate of 57.4%). All assessments were carried out by the first author as a part of his MD thesis. The average time for assessment was between 1 and 1.5 h.

Healthy controls group matched for gender, age and years of education were recruited from the community through convenience sampling. Controls were screened to ensure that they did not have a lifetime diagnosis for any Axis-I psychiatric disorder as per DSM-V (except nicotine dependence), organic brain disorders, intellectual developmental disorder or any pervasive developmental disorder. Of the 181 people approached, 131 healthy controls completed all assessments.

No incentives were provided for participants.

2.2. Measures

In addition to the checklist for socio-demographic profile, baseline clinical information of OCD subjects was collected from the structured proforma used in the OCD clinic. The structured proforma was administered by post-graduate trainees in mental health to all new patients registered at the unit. This diagnostic assessment involves assessing clinical details, symptoms and severity of OCD using Yale-Brown Obsessive Compulsive Scale (YBOCS) (Goodman et al., 1989) and common Axis −1 comorbidities using MINI-6. All patients were clinically interviewed by Senior Consultants attached to the clinic to corroborate the collated information in the structured proforma. All patients with OCD in this study were on treatment with SSRIs at the recommended dose range (Reddy et al., 2017).

2.2.1. Assessments specifically pertaining to the study were the following

2.2.1.1. Impulsivity (Barratt impulsiveness scale). The behavioural construct of impulsivity was assessed for both patients and controls using the self-reported Barratt impulsivity scale (BIS) (JH et al., 1995). The 30 item BIS has good internal consistency and reliability. The BIS has three factors or sub-scales pertaining to, attentional, motor and planning. Assessment were for symptoms of impulsivity and no diagnosis of ADHD was made.

2.2.1.2. Young's internet addiction test- (internet addiction). Young's Internet Addiction Test (IAT) assesses the presence and severity of the dependence on the internet (Young and de Abreu, 2007). The test has 20 items, related to compulsive use and impairment in functioning. Items are scored on a Likert scale ranging from 0 to 5, with scores less than 30 signifying normal levels of usage, 31–49 – mild level of addiction, 50–79 – moderate, and 80–100 signifying severe dependence.

2.2.1.3. Test for mobile phone dependence brief (mobile phone dependence). The test for mobile phone dependence brief (TMD brief) is a 12 item questionnaire which has been validated in the Indian population with a cut-off score of 22 indicating dependence (Chóliz et al., 2016).

2.2.1.4. Sex addiction screening test – revised (Sexual addiction). Sexual "Addiction Screening Test-Revised (SARS-T) screens for the core-components of sexual addiction common across men and woman. It has 20 questions, with a "yes" response to six or more items indicating probable sex addiction (Carnes et al., 2010).

2.2.1.5. Pornography addiction screening tool – (Pornography addiction). The pornography addiction screening tool helps to screen individuals who are at-risk for pornography addiction. The questionnaire has 25 questions, each with 5 potential responses (Bulkley, 2013). The cut-off score of 70 was used indicating individual is addicted to pornography.

2.2.1.6. Compulsive buying scale – (Compulsive buying). Compulsive buying was assessed by Compulsive Buying Scale (CBS) (Faber and O'Guinn, 1992). It consists of 2 items, with the second item consisting of 6 sub-questions. Questions are on a Likert scale from 1 to 5. The cut-off score of < -1.34 is calculated by using a scoring equation.

2.2.1.7. Pathological gambling diagnostic form -DSM-5 (Pathological gambling). The screening questionnaire developed by OASAS (Office of Alcoholism and Substance Abuse Services, New York) used to assess evolving problem and/or at-risk indicators of pathological gambling. The 9-item yes/no questionnaire has been adapted from the DSM V (2013) with 4 or more yes indicating a 'at risk state' (American Psychiatric Association, 2013).

2.2.1.8. Kleptomania symptom assessment scale (K-SAS) (Kleptomania). The K-SAS is an eleven-item self-reporting scale which evaluates shoplifting thoughts, urges and behaviours in the last one week. Each item is scored on a Likert scale from 0 to 4 with a score of >11 indicating serious concern (Grant and Kim, 2002).

2.2.1.9. Pyromania (ICD-10 Criteria). Pyromania was assessed using a screening instrument based on the criteria of ICD-10 (World Health Organization, 1992).

2.2.1.10. Eating attitudes test (EAT-26): (Food addiction). Food addiction was assessed using the 26 item EAT-26 (Garner et al., 1982), on which participants rates themselves of their abnormal eating habits and weight. The instrument is widely used as a screener for capturing any eating disorder with a cut-off score of 20.

2.2.1.11. Exercise dependence Scale – (Exercise addiction). The Exercise Dependence Scale-21 is a 21-item screener which evaluates exercise dependence based on the DSM-IV criteria for substance dependence (Hausenblas and Downs, 2002). Subjects indicate their response to their exercise behaviours in the past 3 months, on a Likert scale with never (1) and always (6). "Individuals are classified into the dependent range if they fulfil 3 or more of the DSM criteria.

2.3. Statistical analysis

Statistical analysis was performed using SPSS version 21. The lifetime prevalence of BA in OCD subjects and healthy controls (HC) were calculated. Those with OCD and HC were compared using symptom severity scores of assessed instruments for BA. Since the scores were not normally distributed, the Mann-Whitney test was used. BAs with very low baseline prevalence were excluded from this analysis.

Further, subjects with OCD were grouped into those who fulfilled the clinical diagnosis of any cooccurring BA and those without. This categorisation was based on a grouping of those who crossed the cut-off scores on any of the BA assessment scales as OCD with BA. The two groups were compared with regards to clinical profile, comorbidity patterns, severity scores, and impulsivity scores (including sub-scores of BIS) using chi-square test and t-tests as appropriate. Total impulsivity scores (BIS-11) were compared across the three groups, OCD, OCD with BAs, and HC using ANOVA. In addition, the correlation between impulsivity scores and individual BAs (excluding BAs with low prevalence) as continuous variables was done for the whole sample. All tests were two tailed with statistical significance at $p < 0.05$.

2.4. Ethics

The study was approved by the Ethical Committee of the Department of Psychiatry, National Institute of Mental Health and Neurosciences, Bengaluru. All patients provided a written informed consent prior to assessment. All participants who screened positive to any of the assessed disorders were psychoeducated and offered appropriate help.

3. Results

A total of 150 OCD patients and 131 healthy controls completed the assessments. Subjects with OCD patients were comparable to healthy controls with regards to demographic profile (Table 1).

OCD subjects reported significantly higher prevalence of atleast one co-occurring behavioural addiction compared to healthy controls [OCD (n-105, 70%); HC (n-76, 58%), $p < 0.036$]. Subjects with OCD when compared to healthy controls reported a significantly higher prevalence of Internet Addiction (29.3% vs 3.1%), pornographic addiction (12.7% vs 0%), and sex addiction (10% vs 1.5%) (Table 2).

When compared to healthy controls, subjects with OCD reported a significantly higher intensity of internet use, mobile phone use, pornography addiction, sex addiction, and food addiction. Compulsive buying showed a trend towards significance (0.055) (Table 3).

When OCD patients with and without BA were compared on clinical characteristics, OCD patients with BA were significantly more likely to be male (71.4% vs 51.1%), with a lower age at onset of OCD (19.90 years vs 26.69 years) and lower age at assessment (28.41 years vs 35.62 years). OCD subjects with co-occurring BAs had significantly higher total impulsivity scores (73.73 vs 66.80), motor sub-scale scores (23.08 vs 18.80) and attentional sub-scale scores on BIS (22.58 vs 19.96). Both groups were comparable with regards to duration and severity of OCD, family history of OCD and comorbidity patterns (Table 4).

Total impulsivity scores were significantly different when compared between HC (55.84 ± 12.80), OCD (66.80 ± 10.25), and OCD with BA (73.73 ± 12.27) ($p < 0.001$) (Fig. 1).

Impulsivity scores when examined in the whole sample, positively correlated with scores of internet addiction, mobile phone dependence, pornographic addiction, sex addiction and food addiction; and negatively correlated with compulsive buying.

Table 1
Comparison of demographic characteristics of OCD patients and healthy controls [N= (150 patients; 131 controls)].

Characteristic	OCD N-150 n (%)	Control N-131 n (%)	x^2/t	p
Age©	30.57 ± 10.26	31.92 ± 10.15	1.106	0.270
Gender				
Male	98(65.3)	86(65.6)	0.003	0.956
Female	52(34.7)	45(34.4)		
Years of Education©	15.75 ± 2.85	14.92 ± 4.15	1.954	0.052

©Mean ± standard deviation.

Table 2
Prevalence of BA among OCD patients (n-150) and healthy controls (n-131).

Behavioral Addictions	OCD N (%)	Control N (%)	x2	p
Internet Addiction	44(29.3)	4(3.1)	34.097	<0.001
Mobile phone dependence	89(59.3)	70(55.4)	0.990	0.320
Pornography Addiction	19(12.70)	0(0.00)	17.797	<0.001
Sex Addiction	15(10.00)	2(1.50)	8.833	0.003
Food Addiction	11(7.30)	11(8.40)	0.110	0.741
Compulsive Buying	10(6.70)	6(4.60)	0.567	0.451
Pathological Gambling	1(0.70)	0(0.00)	0.876	0.349
Pyromania	1(0.70)	0(0.00)	0.876	0.349
Kleptomania	0(0.00)	0(0.00)		
Exercise Dependence	0(0.00)	0(0.00)		

Table 3
Comparison of Intensity of behavioural addictions between OCD patients and Healthy controls[a].

Behavioral Addictions (scores)	OCD Mean ± SD (Median; Q_1-Q_3)	Control Mean ± SD (Median; Q_1-Q_3)	P[a]
Internet Addiction	33.35 ± 25.63 (29; 10.00–54.25)	20.36 ± 13.29 (21; 11–28)	<0.001
Mobile phone dependence	20.55 ± 13.06 (21; 11.75–29.00)	17.64 ± 10.34 (19; 9.00–25.00)	0.045
Pornography Addiction	15.99 ± 21.98 (2; 0.00–30.00)	3.98 ± 8.03 (0; 0.00–4.00)	<0.001
Sex Addiction	1.44 ± 2.88 (0; 0.00–1.00)	0.58 ± 1.35 (0; 0.00–0.00)	0.017
Food Addiction	5.42 ± 8.33 (2; 0.00–2.00)	4.25 ± 8.18 (0; 0.00–0.00)	<0.001
Compulsive Buying	2.19 ± 1.82 (3.34; 1.29–3.34)	2.13 ± 1.48 (3.01; 1.50–3.34)	0.055

[a] Excluded from analysis were the BAs with very low baseline prevalence.

Table 4
Comparison of OCD patients with and without behavioural addictions on clinical characteristics of OCD.

Characteristic	OCD With BA N = 105 n (%)	OCD Without BA N = 45 n (%)	x2/t	p
Age of onset of OCD (years)©	19.90 (8.33)	26.69 (12.64)	3.879	**<0.001**
Age at assessment (years) ©	28.41 (8.58)	35.62 (12.06)	4.153	**<0.001**
Gender -Male	75 (71.4)	23 (51.1)	5.741	**0.017**
Duration Of illness©	101.52 (77.72)	99.09 (83.94)	0.172	0.864
Family History of OCD	20 (19.0)	10 (22.2)	0.198	0.656
YBOCS total score©	24.04 (7.43)	25.49 (6.91)	1.118	0.265
BIS Total Score©	73.73 (12.27)	66.80 (10.25)	3.32	**0.001**
BIS Attentional©$^{\beta}$	*22.58(4.580)*	*19.96(3.908)*	*0.993*	***0.001***
BIS Motor©$^{\beta}$	*23.08(4.801)*	*18.80(3.678)*	*3.418*	***0.000***
BIS Planning©$^{\beta}$	*28.51(6.193)*	*28.24(5.253)*	*0.293*	*0.799*
Psychiatric comorbidities (Present)	46 (43.8)	21 (46.7)	0.104	0.747
Patterns of Co-morbidity				
Depression	31 (29.5)	11 (24.4)	0.403	0.335
Tics	2 (1.9)	2 (4.4)	0.783	0.584
Generalized Anxiety Disorder	3 (2.9)	1 (2.2)	0.49	< 1.000
Panic Disorder	1 (1.0)	1 (2.2)	0.386	0.511
Social Phobia	6 (5.7)	2 (4.4)	0.101	0.751

©Mean ± standard deviation.
BIS – Barratt Impulsivity Scale.
$^{\beta}$ Barrett impulsivity sub-scales.

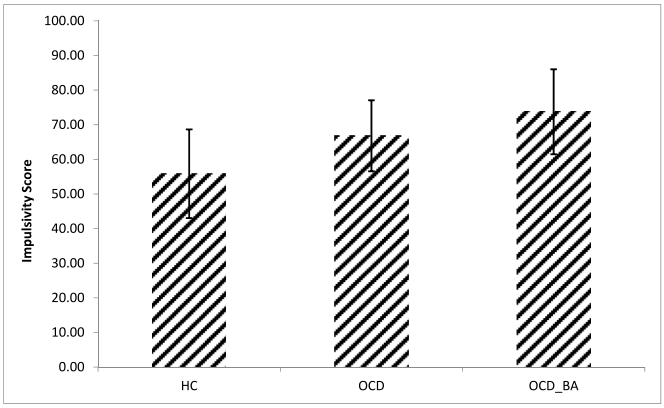

Fig. 1. Comparison of Impulsivity scores between healthy controls (HC), OCD patients (OCD) and OCD with behavioural addiction (OCD_BA) ($p < 0.001$).

4. Discussion

Our study reports that when a comprehensive list of BAs was examined using structured instruments, internet addiction, pornographic addiction and sex addiction were higher among OCD subjects compared to healthy controls. In addition, though not significant as a dichotomous categorical variable, intensity of both mobile phone dependence and food addiction was significantly higher in the OCD group compared to healthy controls. Our hypothesis of higher comorbidity of BAs in OCD has thus been mostly supported as among the BAs examined only compulsive buying did not differ compared to HC. Our study is possibly the first to have compared prevalence using healthy controls group matched for age, sex and education, demographic characteristics with purported influence on prevalence of BAs (Brakoulias et al., 2020)(Bipeta et al., 2015)(Durdle et al., 2008)(Fuss et al., 2019). The higher co-occurrence of internet addiction and compulsive sexual behaviour have been reported previously (Brakoulias et al., 2020)(Bipeta et al., 2015)(Fuss et al., 2019). However, we did not replicate previous findings of higher rates of pathological gambling and compulsive buying among subjects with OCD (Brakoulias et al., 2020)(Lejoyeux et al., 2005)(Durdle et al., 2008). Another study from our centre conducted two decades ago also reported low rates of compulsive buying and pathological gambling in patients with OCD (Jaisoorya et al., 2003). A few BAs like exercise dependence, kleptomania and pyromania appear to be rare among our subjects with OCD. The community prevalence of these disorders has not been examined epidemiologically and it is not clear whether this represents true prevalence or underreporting. This being the case, it needs to be acknowledged there is limited consensus on the concept and criteria of behavioural addictions and assessment instruments (Kardefelt-Winther et al., 2017).

The concept of internet addiction is controversial. Internet is increasingly regarded as a medium for a variety of activities including gaming, gambling, shopping, pornography etc. Additionally, recurrent thoughts of surfing the internet, need to check for information, or a preoccupation to clarify or seek reassurance through online forums could represent a symptom of OCD which may be falsely subsumed under internet addiction (Starcevic and Aboujaoude, 2017). Given the psychopathological overlap, we ensured that all subjects in our study who reported internet addiction were acting on their urges/impulses and experienced pleasure or gratification rather than distress typical of OCD.

Subjects with OCD with BA were more likely to be male, had a younger age of onset of OCD symptoms and a lower age at assessment. The clinical characteristics of the sub-group of OCD with BA have not been examined to date. Being male has been consistently associated with an increased propensity to all addictions, including BAs. Young people, especially students, are more technologically savvy, spend more time online, constantly updating and experience restlessness when offline (Gray, 2018). However, the two groups did not vary with regards duration of illness, symptom severity, family history or the proportion with co-morbidities. Warranting further exploration is, whether the demographic differences reflect the universal risk to BAs or is specific to OCD.

In our study, OCD patients had higher impulsivity scores compared to healthy controls. Further, impulsivity scores on the BIS-11 were higher among OCD patients with BA when compared with those without. Though traditionally OCD is considered compulsive, there is recognition that subjects with OCD have higher impulsivity compared to healthy controls (Benatti et al., 2014). A sub-group of OCD patients have also reported to have impulsive features (Kashyap et al., 2012). However, it has been argued that most studies examining this have reported elevated scores in questions on "cognitive impulsivity" (linked to difficulties in regulating thoughts and distractions) with no group differences in motor impulsivity and non-planning impulsivity (Abramovitch and McKay, 2016). However, our subjects with OCD with co-occurring BAs when compared with those without, had significantly higher BIS-11 'motor impulsivity' scores which taps impulsive behaviour. Further, our finding that impulsivity is positively correlated with all examined BAs (except compulsive buying) across the sample adds to the increasing evidence of the potential role of impulsivity in the development of BAs (Romer Thomsen et al., 2018). Overall, our study finding suggests that in subjects with OCD impulsivity may be trait underlying co-occurring BAs.

It is difficult to speculate on the relationship of BAs and OCD based on co-occurrence, however subjects with OCD may have a higher risk of BAs. For many OCD patients, BAs like internet addictions, pornography etc. may signify relief seeking from the distress of the underlying disease. In subjects with OCD, this relief seeking behaviour may become uncontrollable and compulsive (Bipeta et al., 2015). This is indicated by preliminary evidence of shared neurobiological underpinnings of ritualized behaviour in OCD and BA (Figee et al., 2016; Grassi et al., 2015). Compulsivity in BAs and OCD have overlapping cognitive and affective components. In both, there are engagements in self-destructive repetitive behaviours indicating impaired reward processing linked to involvement of ventral striatum with attenuated dopamine release; diminished ability

to stop suggests cognitive and behavioural inflexibility being related to impaired regulation of ventromedial cortex; and habitual responding indicates imbalances in ventral and dorsal frontostriatal cortex (Figee et al., 2016). Though compulsivity has been implicated, there is substantial evidence for impulsivity being the core process driving BAs (Lee et al., 2019). Impulsivity plays a role in a sub-group of OCD patients (Kashyap et al., 2012). The overlapping features of impulsivity and compulsivity have been the basis of inclusion of both BAs and OCD in the proposed impulsive-compulsive spectrum disorders.

Despite the speculative nature of the overlap, our findings add to existing literature reporting an association of OCD with various BAs. With increasing prevalence of BAs especially the newer technology addictions in the community, this association needs to be given sufficient clinical focus (Chamberlain et al., 2016). Assessments of comorbidity among subjects with OCD should extend beyond the traditional comorbid patterns and include BAs. Since, comorbidity is associated with poorer outcomes, a high index of suspicion is to be maintained (Fineberg et al., 2013). Our study also suggests co-occurrence of BA may signal a unique subset of OCD patients with higher impulsivity traits who may have specific treatment needs.

The strengths of our study include assessment of most currently conceptualised BAs using structured instruments among OCD patients and group matched controls. However, as the criteria of most BAs remain to be defined, the cut-offs represent people with significant risks/concerns and not necessarily a diagnosis. The sample consisted of patients seeking treatment for OCD, hence generalisation is limited beyond the clinical sample. The cross-sectional design precludes any conclusion on longitudinal association or causality. Substance dependence except use of nicotine was an exclusion criterion. However, information on the prevalence and pattern of nicotine use was not explored. Discussion of some BAs like pornography, compulsive buying, gambling is sensitive in conservative societies like India. Hence the true prevalence may be higher given the likelihood of under-reporting during face-to-face interview.

Our findings suggest the need for future research, with prospective studies exploring diagnostic stability, course and outcome, and neurobiological studies to improve our understanding on etiological underpinnings of this relationship. This will help ensure more efficacious interventions for patients with both OCD and BAs.

5. Conclusion

The co-occurrence of BAs among subjects with OCD is not uncommon. Our study suggests that OCD subjects with co-occurring BAs were more likely to be male, with a lower age at onset, lower age at assessment and higher impulsivity. Given the increasing reporting of BAs, clinicians need to aware of the need to explore this in subjects with OCD.

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Author's contribution

Dr Jaisoorya TS, Dr Janardhan Reddy YC, Dr Janardhanan C Narayanaswamy and Dr Shyam Sundar A designed the study and wrote the protocol. Dr Dev Rai carried out the data collection. Dr Dev Rai and Dr Jaisoorya TS wrote the first draft. Dr Janardhan Reddy YC, Dr Janardhanan C Narayanaswamy and Dr Shyam Sundar A made modifications on the first draft. All authors contributed to and have approved the final manuscript.

Ethics

The study was approved by the Institutional review board of NIMHANS, Bengaluru. All patients provided a written informed consent prior to assessment. All participants who screened positive to any of the assessed disorders were psychoeducated and offered appropriate help.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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